Debug PRU with Code Composer Studio

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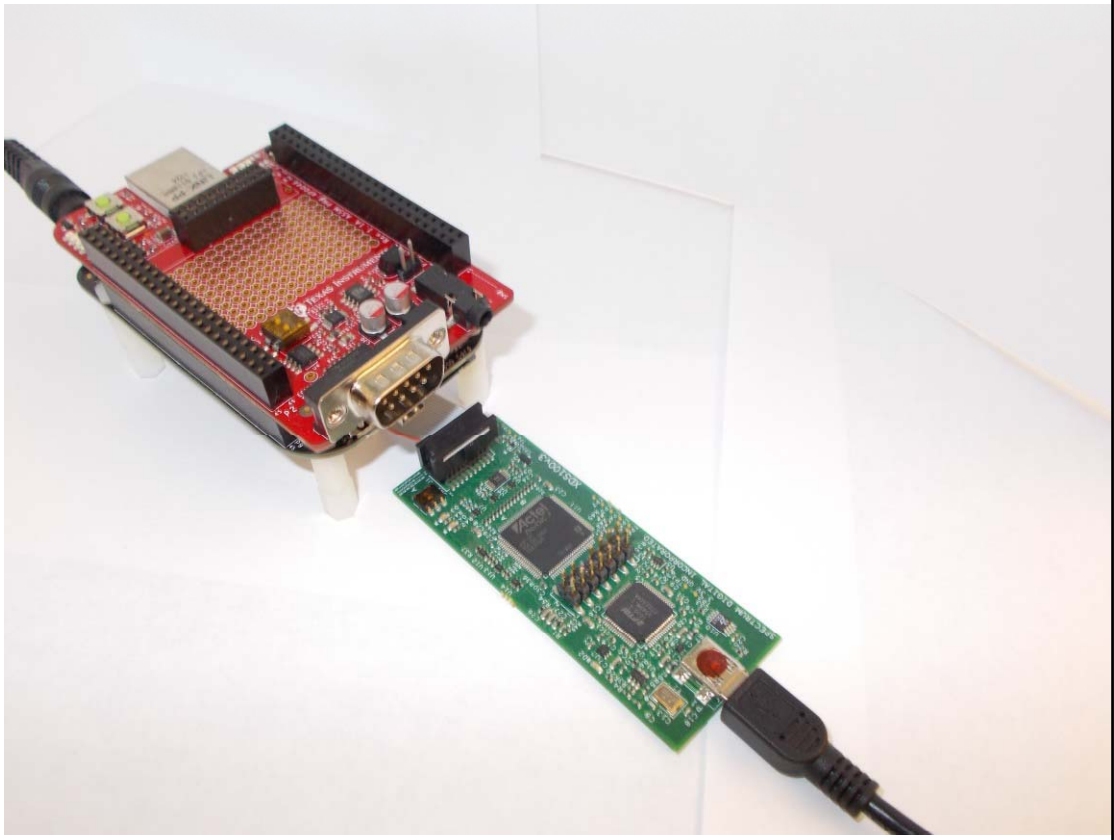
Hardware: BeagleBone Black

- BeagleBone Black (BBB)
  - AM335x device
  - 2 PRU's embedded in AM335x
  - Add JTAG header to P2 on back
    - Optional standoffs
- Debug Probe
  - XDS100 or XDS200
  - Screen shots show XDS100v3
- PRU Cape
  - Add-on board for learning about PRU
- BBB only info is marked

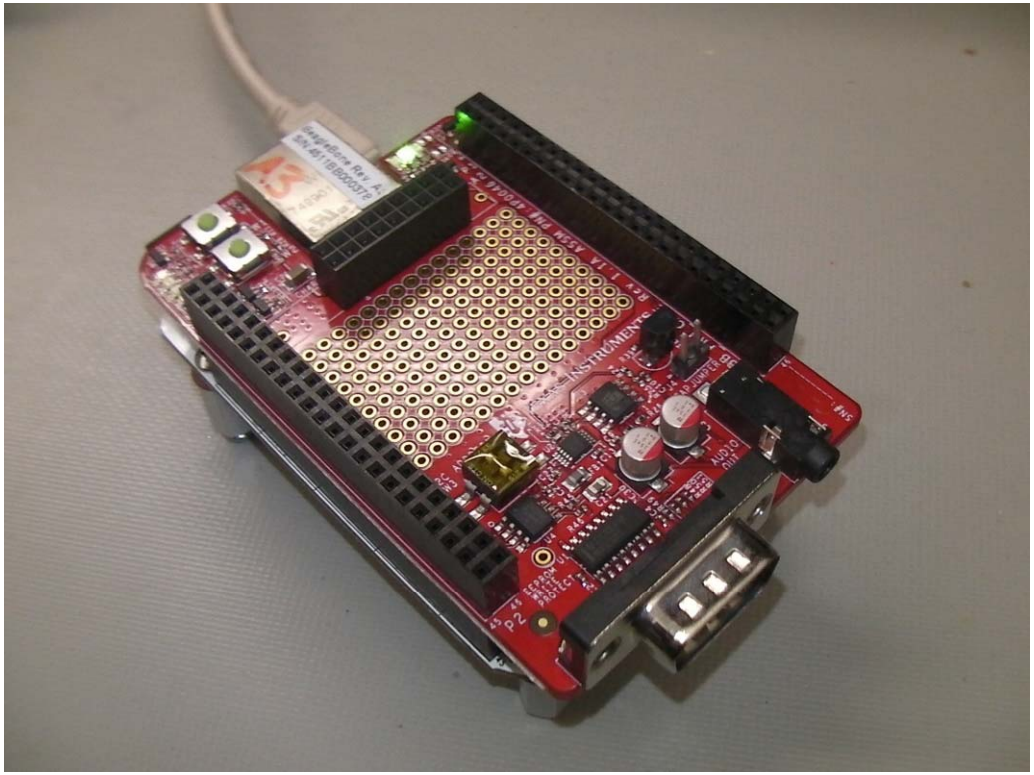
BBB Only



Hardware: BeagleBone White

- BeagleBone White (BBW)
  - AM335x device
  - 2 PRU's embedded in AM335x
  - More expensive than BBB
- Debug Probe
  - XDS100v2 built-in
- PRU Cape
  - Add-on board for learning about PRU
- BBW only info is marked

BBW Only



Lab Conventions

- Lab steps are in black and numbered for easier reference
  1. …
  2. …
- Descriptive text is written in blue

Install Software

- CCS version 6.1
- Install PRU C compiler
  1. Select *View→CCS App Center*
  2. To right of **Code Composer Studio Add-ons**, click on **See more >**
  3. Select **PRU Compiler**
     - Version 2.1.0 or later
  4. Click on **Install Software**
     - In the upper left corner. Scroll up if needed.

Workaround: Avoid Linux on BBW

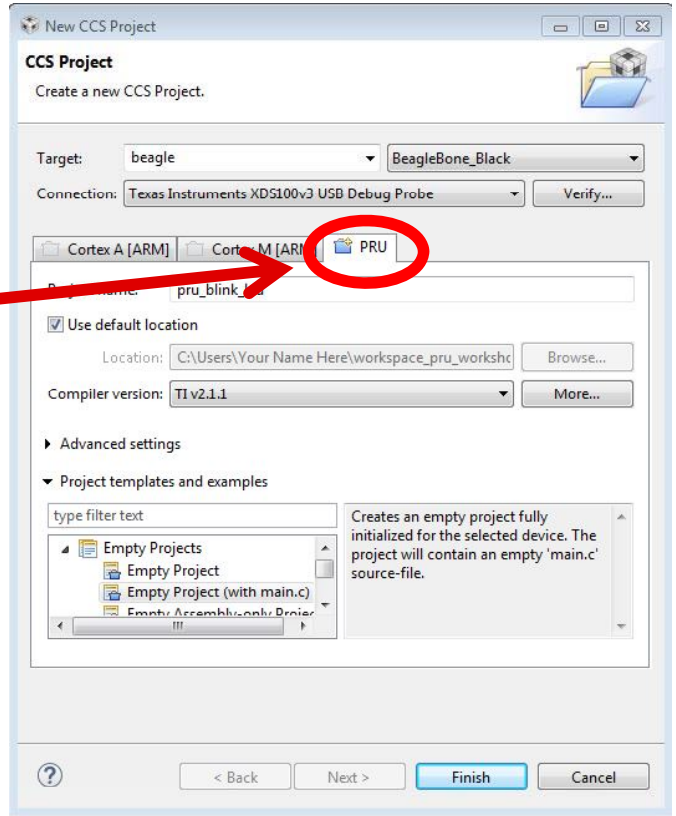
- Prevent Linux from running
- Eject micro SD card before power up
- OK for Linux to run on BBB

BBB: Create Blink LED Project

BBB Only

From CCS Edit perspective

1. Go to *File→New→CCS Project*
2. Set **Target** to *BeagleBone_Black*
3. Set **Connection** to your debug probe
4. Select **PRU** tab
5. Enter project name
   - This workshop uses *pru_blink_led*
6. Leave **Project templates and examples** at default
   - Empty Project (with main.c)
7. Click **Finish**
   - CCS creates the project

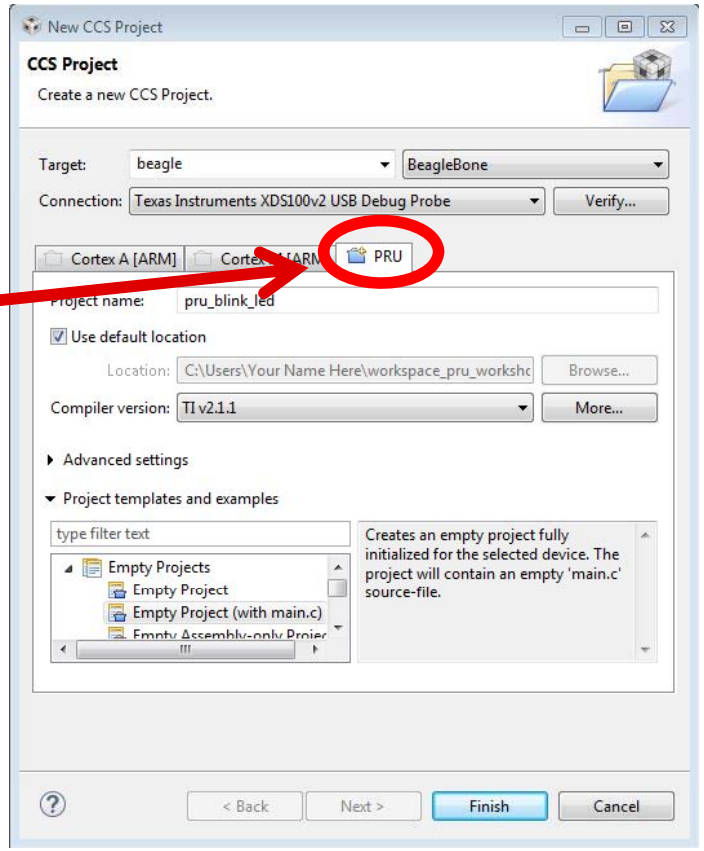


BBW: Create Blink LED Project

BBW Only

From CCS Edit perspective

1. Go to *File→New→CCS Project*
2. Set **Target** to *BeagleBone*
3. Set **Connection** to XDS100v2 USB
4. Select **PRU** tab
5. Enter project name
   - This workshop uses *pru_blink_led*
6. Leave **Project templates and examples** at default
   - Empty Project (with main.c)
7. Click **Finish**
   - CCS creates the project



Replace All Code in main.c with These Lines

```
#include <stdint.h>    // for uint32_t
volatile register uint32_t __R30;

int main(void)
{
        uint32_t gpo;

        while (1)
        {
                gpo = __R30;
                gpo ^= 0xF;
                __R30 = gpo;
                // 10 to the 8th == half-second delay
                __delay_cycles(100000000);
        }
}
```

Build

1. Save changes to main.c with control-S
2. Click build icon

Correct any errors found.

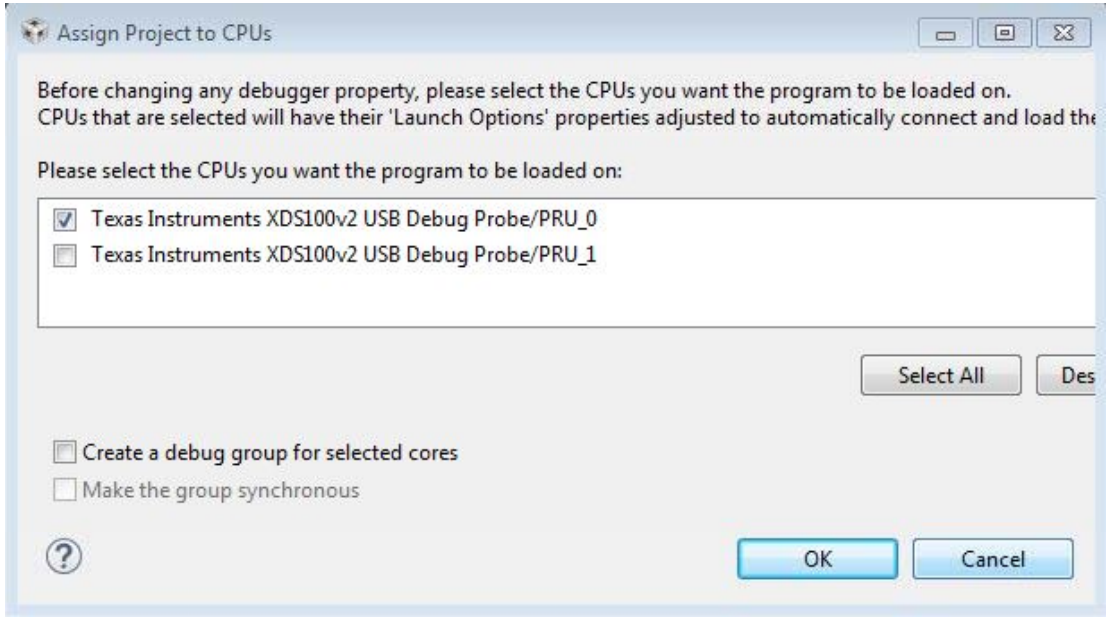
Debug Configuration Described

- It is tricky to access a deeply embedded CPU on a complicated SOC
- When you start a debug session, CCS automatically creates a *Debug Configuration*
  - Contains detailed info about how to start up the debugger part of CCS
- Accessing the PRU requires adding a special script to the Debug Configuration

Create Initial Debug Configuration

1. Right click the project name and choose **Properties**
2. In left pane highlight **Debug**
3. CCS asks which PRU core to load. Check *PRU_0*. Clear *PRU_1*.
4. Click **OK**
5. Click **OK** again

Adding the script is next.



Add PRU Startup Script to Project

- More initialization needs to happen when debugger launches
  - Some on Debug Access Port (DAP), some on PRU
- Cannot perform with GEL
  - GEL script can only be associated with one CPU
- Can control from a Debug Server Scripting (DSS) script
  - Can perform operations on all CPU's

1. Download example DSS script
   - http://processors.wiki.ti.com/index.php/Debug_Configuration_Initialization_Scripts
   - File named bbb_pru_startup.js
2. If Project Explorer view is not present, select *View→Project Explorer*
3. Paste bbb_pru_startup.js into pru_blink_led project next to main.c
   - CCS Project Explorer view works just like Windows Explorer

Add Startup Script to Debug Configuration

1. Click down arrow next to Debug icon and select **Debug Configurations**

2. In left pane, expand *Code Composer Studio* entry
3. Highlight the *pru_blink_led* configuration
4. Next to **Initialization Script** click **Workspace**
5. Open tree for *pru_blink_led* and select *bbb_pru_startup.js*
6. Click **OK**
7. Click **Apply**
8. Compare with screen shot in next slide
9. Click **Close**

This DSS script runs every time debugger is launched.

Debug Configuration

Debug Configurations

Create, manage, and run configurations

Start CCS Debug launch

type filter text

C/C++ Remote Apl

Code Composer St

pru_blink_led

Launch Group

Filter matched 4 of 4 items

Name: pru_blink_led

Main | Program | Target | Source | Common

Use default target configuration

Target Configuration ${target_config_active_defa | File System... | Workspace...

Initialization Script C:\Users\Your Name Here\\ | File System... | Workspace...

M3_wakeupSS
CortxA8
PRU_0
PRU_1

CPUs
Non Debuggable Devices
Routers

Synchronize the properties for all compatible CPUs

Apply | Revert

Debug | Close

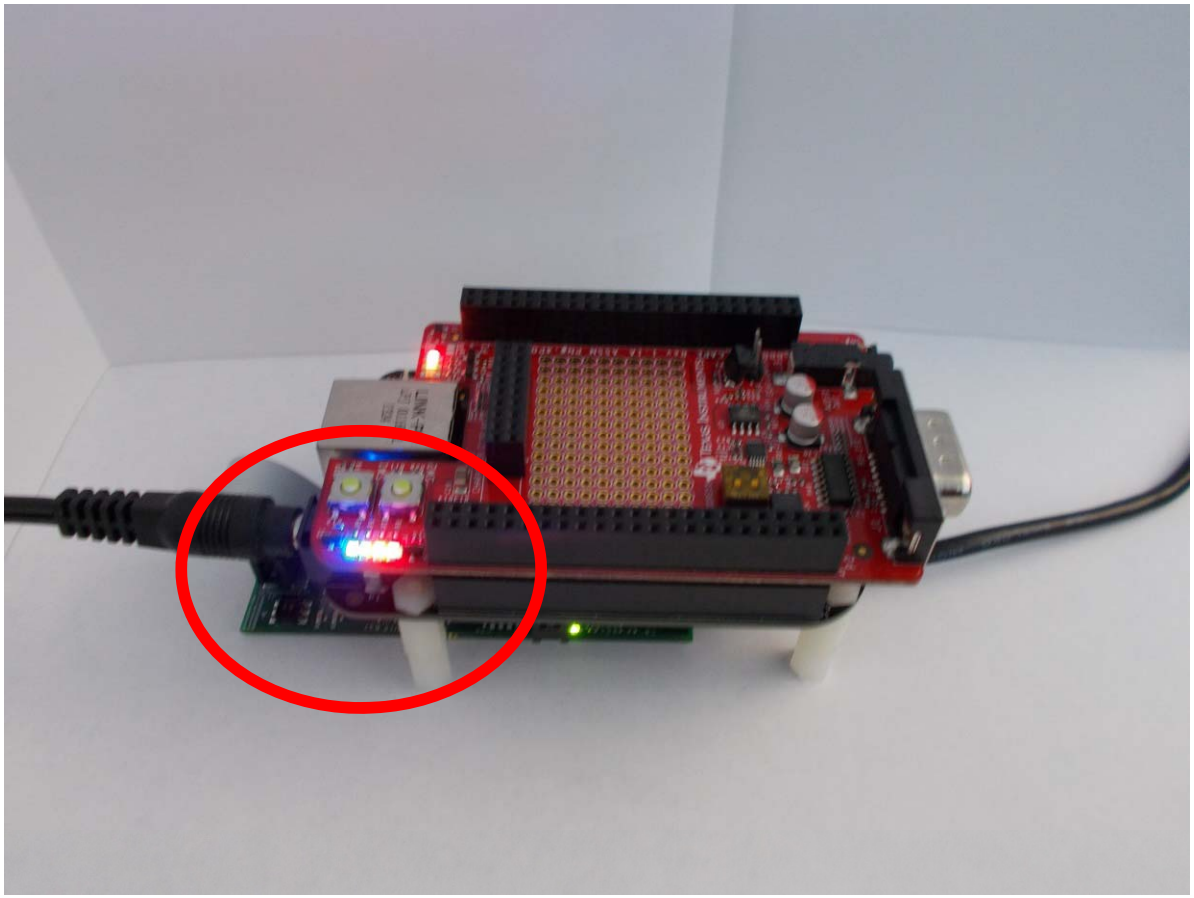
Execute the Code

1. Click Debug icon
2. Click Resume Execution icon
3. See the LED's blink!
4. Click Suspend to halt execution

Ignore any error message about: invalid register index 8

This is a known issue which will be fixed.

Blinking LED



For More Information

- E2E Forum
  - http://e2e.ti.com
  - Community forum. Search, ask, answer questions.
- TI Processors Wiki
  - http://processors.wiki.ti.com
  - Technical articles on practical topics
- PRU Software Support Package
  - http://www.ti.com/tool/pru-swpkg
  - Framework and examples for developing PRU software